Statistical Analysis Plan

A Randomized Clinical Trial Comparing the Overall Adverse Event Rate of Inguinal Hernia (IH) Repair Prior To NICU Discharge Versus IH Repair After NICU Discharge and Beyond 55 Weeks Post Menstrual Age in Premature Infants.

Short Title: **H**ernia **i**n **P**remies (HIP Trial)

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1. Overview

The purpose of this Statistical Analysis Plan (SAP) is to provide all relevant information of the planned analyses for the HIP Trial as specified in the study protocol and to clarify any potential points of confusion in the original protocol. The planned analyses identified in this SAP will be included in future manuscripts as deemed necessary by the HIP subcommittee. It is not the intent of this SAP to identify all possible analyses that may be completed using the HIP database. The purpose of this SAP is only to describe the analyses required to address the objectives/specific aims/hypotheses explicitly stated in the protocol. Additional exploratory analyses not identified in the protocol or not included in this SAP may be performed by the HIP subcommittee.

2. Study Background and Rationale

The HIP Trial is a multicenter randomized clinical trial (RCT) conducted to determine whether early inguinal hernia (IH) repair (prior to neonatal intensive care unit [NICU] discharge) or late inguinal hernia repair (>55 weeks post menstrual age and after NICU discharge) is the safer surgical approach for premature infants who are diagnosed with an inguinal hernia while in the NICU. Because the timing (and duration) of general anesthesia is believed to be an important determinant of neurodevelopmental outcomes beyond infancy, the HIP trial will also compare neurodevelopmental outcomes, as assessed by the Bayley Scales of Infant Development, 3rd edition (BSIDIII), at 2 years corrected age, for infants in the early versus late IH repair groups.

3. Study Hypotheses and Objectives

Primary objective

To compare the overall safety of early and late IH repair by comparing the proportion of infants with ≥1 serious adverse event, measured from enrollment through 10 months later.

Secondary Objectives

1. To compare the number of hospital days from enrollment through 10 months beyond enrollment between early and late IH repair, as a measure of the impact of adverse events and resource utilization between the two groups.

2. To compare the safety of early and late IH repair with respect to neurodevelopmental outcome at 22-26 months corrected age by comparing the cognitive composite BSIDIII scores between groups.

3. To assess differences between early and late IH repair groups with respect to secondary safety outcomes.

4. Using Bayesian and frequentist analyses, evaluate specific hypotheses about baseline patient characteristics that may influence the relative safety of early and late IH repair.

Primary Hypothesis

Late IH repair (> 55 weeks post-menstrual age [PMA], which is expected to be ~5 months after NICU discharge in most cases) in premature neonates with an IH will result in a 10% absolute reduction in the proportion of infants with ≥ 1 serious adverse event (SAE) compared to early IH repair (prior to NICU discharge, ~ 38 weeks PMA). We estimate 30% of infants in the early repair group will have at least one SAE compared to 20% in the late IH repair group.

Secondary Hypotheses

1. Late IH repair will be associated with a 3-day reduction in the median total number of hospital days during the study observation period (i.e., enrollment until 10 months later) compared to early IH repair (8 total hospital days for early IH repair versus 5 for late repair).

   **Clarification of inconsistency in final trial protocol: in the original trial protocol, the expected median total number of hospital days was listed as 18 for early IH repair and 15 for the late repair group (page 3), which is incorrect. The correct values were presented on page 12 of the final protocol and are as written here.

2. Late IH repair will be associated with a 0.4 SD increase in mean scores on the BSIDIII Cognitive Composite score (6 points absolute difference) compared to early IH repair.

3. Despite an *overall* treatment effect favoring late IH repair with respect to SAEs, the actual treatment effect will vary for pre-defined and specified patient subgroups defined in the trial protocol ("treatment effect modifiers" section, page 13).

4. Trial Methods

4.1 Trial Design

The HIP trial is a randomized, multicenter parallel-group trial. Infants are randomized 1:1 to early or late IH repair.

4.2 Treatment Groups

a. Early IH Repair: IH repair is performed after randomization, when the neonatology team believes the infant is optimized from a medical perspective, and prior to NICU discharge; ~38 weeks PMA, based on pilot trial and experience in practice.

b. Late IH Repair: IH repair will be performed when infant is > 55 weeks post menstrual age and after NICU discharge. This is estimated to typically be approximately 5 months after NICU discharge. This PMA was selected as the goal time for late repair because most children's hospitals consider allowing the operation to be done as an outpatient at this time and this reflects current surgical care. The decision regarding admission will be determined by the providers for each infant according to site guidelines.

4.3 Randomization

Randomization is stratified by site and GA stratum (<28 weeks and ≥28 weeks estimated GA) using permuted variable block size. Randomization occurred 24 hours a day, 7 days a week through a REDCap electronic data capture system. In cases of multiple gestations qualifying for inclusion the first twin was randomized in the usual manner and the sibling was assigned to the same treatment group.

4.4 Sample Size

To test the primary hypothesis that overall SAE rates will be 30% in the early repair group compared to 20% in the late repair group 293 infants per group (N = 586 total) are required to have 80% power (two-sided alpha of 0.05) to detect a 10% absolute difference. With a 95% follow-up rate, we plan on randomizing 615 infants. For the secondary outcomes, we will have 98% power to detect a median difference of 3 hospital days (assuming median=8, mean=18 for early group and median=5, mean=13 for the late group and a log normal distribution). For BSIDIII scores, we will be able to detect a 0.4 SD difference (6 points on the BSIDIII) assuming 200 infants are assessed at 22-26 months.

4.5 Interim Analysis

One formal statistical interim analysis was planned on the primary endpoint when 10-month data was available for 200 randomized infants. Further details of this analysis and stopping guidelines are provided in the study protocol (Page 14).

4.6 Timing of final analysis

Final analysis of the early vs late IH repair comparison was planned to take place when 10-month outcome data was available for 586 randomized infants. However, in April 2021, the HIP trial stopped enrollment after an interim analysis found a 97% probability of decreased rate of SAEs in the late group. This probability crossed the pre-specified efficacy stopping threshold of 95%. Consistent with our overall approach, the interim analyses were primarily Bayesian.

4.7 Timing of outcome assessments

Primary outcome is assessed at 10 months post-randomization. Neurodevelopmental outcomes are collected at 22-26 months corrected age. More details are presented in the study protocol.

5. Statistical Principles

5.1 Analysis framework and reporting procedures

Complementary frequentist and Bayesian analyses will be conducted for the primary outcome (proportion of children with ≥ 1 SAE) and important secondary outcomes of death, total hospital days, and BSIDIII scores. Conduct and reporting of both frequentist and Bayesian results has been previously recommended to provide a thorough presentation of the evidence of an intervention effect.[1] While frequentist analyses are typically reported for RCTs of medical interventions, the results are often erroneously dichotomized into "significant" and "non-significant" based only on the p-value. Importantly, frequentist statistics cannot provide an important quantity, the direct probability of treatment benefit or harm from an intervention. Bayesian statistics calculate this probability that can be directly used by clinicians and parents when deciding on treatment options and for these reasons we consider our Bayesian analyses to be primary and frequentist analyses to be secondary. This important emphasis has been made prior to the conduct of any final analyses. Our investigator group has been involved in several RCTs using this approach and have been influenced by other recent publications.[2-5]

We will report estimates of treatment effect (relative risk and risk difference, group mean differences) and 95% confidence intervals (CI) for all frequentist analyses. For Bayesian analyses, we will report posterior medians, 95% credible intervals (CrI), and probabilities of any intervention benefit/harm and probability of a clinically important intervention effect.

5.2 Protocol Deviations

Protocol deviations were collected using a specified set of descriptions. Possible values for protocol deviations were:

- Study intervention not completed at specified time (either early or late IH repair)
- Wrong treatment intervention applied
- Infant ineligible
- No consent
- Wrong stratification variables used at randomization
- Other violation

Protocol deviations will be descriptively summarized in terms of infants per arm with any deviations and in terms of number of deviations per arm.

5.3 Analyses Populations

We will conduct analyses in two study populations, a modified intention-to-treat (mITT) and per-protocol (PP) populations. The mITT analysis dataset will include all randomized infants with ascertained primary outcomes. The PP analysis dataset will include all infants who received their randomized intervention (early or late IH repair) and have a determined primary outcome.

6. Trial Population

6.1 Screening population

All infants born at < 37 weeks gestation with a diagnosis of IH at a participating site are screened for inclusion in the trial. Total number of eligible patients, number of eligible not enrolled, and reason for non-recruitment will be reported.

Eligibility

Inclusion Criteria: 1) GA < 37 weeks, 0 days; 2) In a NICU / other nursery, at participating sites; 3) Diagnosed with an IH per the pediatric surgeon; and 4) Parental consent and providers willing to randomize the infant.

Exclusion Criteria:

1. Associated factor that impacts timing of repair (e.g. clinical factors that might preclude early IH repair)

2) Infant is undergoing another operative procedure and IH repair is planned as a secondary procedure such that timing of IH repair is impacted (e.g., fundoplication or gastrostomy tube is planned, and IH repair is considered a secondary procedure)

3) Known major congenital anomaly or chromosomal abnormality

4) Family unable to return for follow up and late IH repair or likely unable to monitor IH as outpatient (e.g., maternal incarceration or moving away from the area).

A CONSORT flow diagram will be used to summarize the number of patients who were: eligible and randomized; eligible but not randomized (reasons will be presented); received the randomized intervention; did not receive the randomized intervention; lost to follow-up; discontinued the intervention; randomized and included in the primary analysis; randomized and excluded from the primary analysis (reasons will be presented).

6.2 Withdrawal / Follow-up

- Infants can be lost-to-follow-up between NICU discharge and the time of their scheduled repair (for late IH repair) or their 10-month follow-up phone call. Infants in centers with neurodevelopmental assessments can also be lost between 10-month and 22-26 month follow-up assessment.
- Infants who withdraw from the study will not have any data collected beyond the timepoint of withdrawal. Data collected prior to withdrawal will be available for summarization and analysis.
- Numbers of infants who withdraw or are lost to follow-up will be presented in the CONSORT diagram.

6.3 Baseline Patient Characteristics

The following baseline characteristics are collected and will be summarized for all infants randomized into the trial:

- Infant baseline characteristics
  - Age at randomization (gestational, postmenstrual, chronological)
  - Sex
  - Race
  - Hispanic ethnicity
  - Birthweight (grams)
  - Singleton gestation
  - Inborn/outborn status
  - 1-minute Apgar scores
  - 5-minute Apgar scores
  - Number of surgeries before randomization
  - Apnea or bradycardia pre-enrollment
  - Mechanical ventilation >24 hours prior to randomization
  - Intraventricular hemorrhage? If yes, highest grade and laterality
  - Comorbidities – ROP, RDS, BPD, CLD, cardiac anomaly, NEC, SIP
  - Inguinal hernia reducibility difficulty prior to enrollment
  - Receiving caffeine or steroids?
  - Respiratory status

- Maternal baseline variables
  - Mother’s age
  - Mother’s race
  - Ethnicity
  - Marital status
  - Education status
  - Parity
  - Multiple Birth
  - Prenatal care
  - Medical insurance
  - Zip code of primary residence

Categorical variables will be summarized by numbers and percentages. Continuous data will be summarized by mean, SD and range if normally distributed and median, IQR, and range if data are skewed.

7. Statistical Analysis

7.1 Primary Outcome

The primary outcome is whether an infant experienced any SAEs from randomization to 10-month post-randomization. SAEs recorded are shown in Table 3 from the protocol (replicated below). Specific definitions (based on standard definitions when available) are listed in the study protocol Appendix for all measured SAEs. All final recorded SAEs were reviewed and adjudicated by the primary outcome adjudication committee as described in the protocol (page 11).

Table 3. Serious adverse events to be identified and tabulated in HIP Trial

Cardiorespiratory or Anesthesia – related SAE	Inguinal Hernia or Surgery – related SAE
Death	IH incarceration
Use of extracorporeal membrane oxygenation (ECMO)	IH strangulation
Cardiac arrest	IH recurrence
Cardiopulmonary resuscitation	Reoperation for IH or related to initial operation
Hypotension treated with vasoactive drugs	Injury to adjacent structure / organ
Apnea treated with increased respiratory support	Wound disruption (deep)
Local anesthetic toxicity (e.g. seizure)	Surgical site infection (deep)
Prolonged intubation / ventilation (>48 hours post op)	Other SAE involving intervention to prevent or address patient harm
Unplanned reintubation	
Intraoperative unplanned extubation	

7.2 Secondary Outcomes

- Highest level of harm: assigned by adjudication committee for patients that have had ≥ 1 SAE according to the National Coordinating Council for Medication Error Reporting and Prevention (NCC MERP) Index
- Total number of hospital days: number of days from randomization to NICU discharge, readmission hospital days, hospital days post-surgery.
- Prolonged hospital stay: >30 days post-surgery
- NICU length of stay
- Post-surgery length of stay
- IH Repair occurred (allows assessment of no IH repair being done for various reasons)
- BSIDIII cognitive composite score
- Overall adverse event rates (any AE)

7.3 Analysis Methods

All outcomes will be summarized by treatment group. Categorical outcomes will be summarized by numbers and percentages. Count outcomes will be summarized by median, IQR, mean, and SD. Continuous normal outcomes will be summarized by mean and SD.

Multilevel regression models will be used to analyze all outcomes. For each outcome, the same model will be used for the frequentist and Bayesian analysis. Binary outcomes will be analyzed with a logistic model. Ordinal outcomes will be analyzed with a proportional-odds logistic model. Count outcomes will be analyzed with a negative binomial model. Continuous outcomes will be analyzed with a normal regression model. All models will include a random intercept for site and will include GA group (< 28 weeks and ≥ 28 weeks estimated GA) as a covariate.

We will report relative risks (RR) and risk differences (RD) for binary outcomes (calculated from logistic model), RRs for count outcomes, odds ratios (OR) for ordinal outcomes, and mean group differences for continuous outcomes. We will report 95% CIs (95% CrIs for Bayesian results) for all outcomes. For probabilities of benefit/harm, we will report probabilities of *any effect* and probabilities of at least 3% and 5% reduction in the risk for SAE. Although we hypothesized a 10% reduction in the proportion of infants in the late IH repair group, most stakeholders would likely agree that a smaller reduction (3 or 5%) is above the minimal clinically important difference.

All analyses will be conducted in R. Bayesian models will be fit using Markov chain Monte Carlo (MCMC) methods implemented in Stan via R packages 'rstanarm' and 'brms'. Each model will be fit using three MCMC chains. Each chain will involve at least 5,000 burn-in samples and at least 50,000 additional iterations.

7.3.1 Bayesian prior specifications

Bayesian analyses will use neutral priors for all outcomes, since there has never been a previous RCT comparing these two treatment options. For categorical outcomes, we will assume a prior centered at OR of 1.0 with 95% CrI of 0.2-4 for the intervention effect. This will take the form of a Normal distribution with mean of 0 and SD of 0.7 in the log OR scale. For count outcomes, the prior for the intervention effect will be centered at RR of 1.0 with 95% CrI of 0.33-3.3. For the intercept we will use a vague Normal(0, SD=10) distribution and for GA group a weakly informative Normal(0, SD=1) prior. For the standard deviation of the random intercept, we will use a half-Normal(0, SD=1) prior distribution.

For continuous outcomes, we specify priors for standardized variables. For the intervention effect we will use a Normal(0, SD=2) prior. For the intercept and GA group we will use Normal(0, SD=10) and Normal(0, SD=3) priors, respectively. For the standard deviation of the random intercept, we will use a half-Normal(0, SD=2) prior distribution.

7.3.2 Assumption Checks

Assumptions of frequentist models will be evaluated using residual plots.

For Bayesian models, convergence of MCMC chains will involve visual comparisons of the trace plots from the 3 chains generated for each model. Quantitative checks such as the Geweke and Gelman/Rubin tests will be completed to ensure convergence to the posterior distribution.

For Bayesian models, posterior predictive checks will be used to determine if the resulting model mimics the data sufficiently well for the inferences that will be made from this trial. The posterior predictive check will involve simulating 1,000 trials from the posterior predictive distributions of the model parameters and summarizing the average probability/number of counts/mean of the outcome by:

- Treatment
- GA group

When the above 1000 simulated trial averages are plotted by the above categories and the observed proportions are overlayed on the plots, then the observed proportions should lie within the cloud of points generated by the simulations and not appear to be outliers.

7.3.3 Sensitivity Analysis

In addition to adjusting for GA group and center as a random effect, sensitivity analyses will be conducted if important baseline covariates are imbalanced across the two intervention groups. This analysis will only be completed for the primary outcome and will be based on the mITT sample.

For the primary outcome, in addition to the mITT sample-based analysis, a PP analysis will be done using the PP sample.

7.3.4 Subgroup Analyses

The primary outcome will be reanalyzed for subgroup exploratory analyses. These analyses will be completed using the mITT sample. To investigate treatment effect modifiers for the primary and secondary outcomes (total hospital days and BSIDIII scores), we will use Bayesian hierarchical models with terms of interaction between treatment group (early, late) and the prespecified potential moderators: GA ($<$28; $\geq$28 weeks) bronchopulmonary dysplasia (yes/no), maternal education $\leq$12 years (yes/no), prenatal care (yes/no), IH difficult to reduce (yes/no). The conservative Bayesian approach of Dixon and Simon[6] allows us to specify a priori how likely (or unlikely) it is for subgroup differences to be present and to shrink the subgroup estimates to the overall mean treatment effect.

Prior distributions for main effects will be the same as for the primary analysis. A Normal(0, SD=0.6) will be used for interaction terms. Point estimates of treatment effect and 95% credible intervals for each subgroup will be reported along with probability of treatment benefit/harm. We will conduct similar subgroup analyses to examine differences in treatment effect according to gender, race/ethnicity, although such differences are not expected (required for NIH Phase III RCTs).

Two additional exploratory subgroup analyses will be performed (not stated in trial protocol): the impact of laparoscopic versus open IH repair on primary and secondary outcomes has become more important given the increasing use of laparoscopic approach to these operations. The distance of the primary residence of the family to the treating facility will also be evaluated to investigate whether longer distance from the facility might impact the treatment effect by favoring early repair.

7.3.5 Analyses of association between gestational age and risk of SAE

In addition to the planned subgroup analyses by GA group, we will also conduct analyses to investigate the effect of GA on the rate of SAE. We will treat GA as a continuous variable in the model and using splines or polynomials we will evaluate whether a linear or non-linear relationship provides the best fit for the data. We will use splines or polynomial to model the association. All other model details will be as described for the primary analysis.

We will then evaluate an interaction between the final GA term (linear or non-linear) and intervention effect for the primary outcome using the same methods described in the Subgroup Analyses section.

7.3.6 Ad-hoc Analyses

Other ad hoc analyses may be done in addition to those specified here at the primary and secondary papers are developed. These will be clearly labeled in any publication as ad hoc analyses motivated by our findings as we completed the planned data analyses.

7.4 Missing Data

No imputation for missing data will be done.

7.5 Statistical Software

All analyses will be conducted in R.

References

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